

PRE-RESULTS PREVIEW - NOT FOR SUBMISSION. Scientific result fields and author declarations remain unresolved. Do not upload this file to a journal.

CLAIM 2024 checklist — Q1/Q2 v2 study

Guideline: Tejani AS, Klontzas ME, Gatti AA, et al. *Checklist for Artificial Intelligence in Medical Imaging: 2024 Update*. Radiology: Artificial Intelligence. 2024;6(4):e240300.

doi:10.1148/ryai.240300.

This is a pre-results compliance template. “Yes” and “Partial” entries remain pending until real results are rendered and final page/line references are inserted. Repository paths identify implementation evidence; they do not replace reporting in the manuscript. Author-dependent declarations are not inferred from code.

ID	CLAIM 2024 item	Status before final layout	Final manuscript location	Repository evidence / explanation
C01	Identify the work as an AI methodology study and name the technology in the title or abstract	yes_pending_final_page_lines	[FINAL PAGE/LINES: title and abstract]	manuscript/q1q2_v2_manuscript.template.md; deep-learning segmentation and model families are stated without architectural novelty claims.
C02	Provide a structured summary of design, methods, results, conclusions, population, partitions, and availability	partial_pending_final_results	[FINAL PAGE/LINES: structured abstract]	The template covers design and methods; result values and availability identifiers remain Gate J/K outputs.
C03	State scientific and clinical background, intended use, and role of the AI approach	yes_pending_final_page_lines	[FINAL PAGE/LINES: Introduction]	Intended role is methodological segmentation evaluation; no clinical deployment claim is made.
C04	State aims, objectives, and hypotheses	yes_pending_final_page_lines	[FINAL PAGE/LINES: Introduction]	The capacity-controlled RES question and prespecified secondary analyses are explicit.
C05	State whether the study is prospective or retrospective	yes_pending_final_page_lines	[FINAL PAGE/LINES: Study Design]	Retrospective public-data methodology study.
C06	Describe the study goal and predictive task	yes_pending_final_page_lines	[FINAL PAGE/LINES: Study Design and Models]	Four-class multimodal MRI segmentation with regional and lesion-level evaluation.

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C07	Describe data sources and their alignment with intended use	yes_pending_final_page_lines	[FINAL PAGE/LINES: Development and External Cohorts]	BraTS 2020 development and BraTS-Africa external testing; reports/q1q2_v2/external_dataset_selection.md.
C08	State inclusion and exclusion criteria	yes_pending_final_page_lines	[FINAL PAGE/LINES: Development and External Cohorts]	Cohort roles and exclusion of other neoplasms from confirmatory inference are explicit.
C09	Describe preprocessing in reproducible detail	yes_pending_final_page_lines	[FINAL PAGE/LINES: Preprocessing]	configs/data/preprocessing_pilot_cached.yaml; fold and external boundaries are declared.
C10	Describe any selected subset, crop, or extracted image portion	yes_pending_final_page_lines	[FINAL PAGE/LINES: Preprocessing]	Training sampling may enrich tumor content; complete validation/external volumes are retained.
C11	Describe de-identification	partial_pending_final_results	[FINAL PAGE/LINES: Data Availability and Ethics]	Public deidentified sources are used; source-specific de-identification is not repeated locally. Final ethics language requires author confirmation.
C12	Describe handling of missing data	yes_pending_final_page_lines	[FINAL PAGE/LINES: Cohorts and Statistical Analysis]	Modality completeness, metadata missingness, subgroup missingness, and failed predictions have explicit dispositions.
C13	Report relevant image-acquisition information	partial_pending_final_results	[FINAL PAGE/LINES: Cohorts and Supplement]	Available scanner, field-strength, spacing, geometry, and resolution metadata are reported; unavailable acquisition fields remain missing rather than imputed.
C14	Define the reference standard	yes_pending_final_page_lines	[FINAL PAGE/LINES: Cohorts and Central Evaluation]	Source-provided tumor subregion labels and exact WT/TC/ET mapping are stated.
C15	Explain the rationale for the reference standard	yes_pending_final_page_lines	[FINAL PAGE/LINES: Cohorts and Limitations]	BraTS-compatible labels support the frozen task while annotation uncertainty remains a limitation.

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C16	State the source of reference annotations and annotator information	partial_pending_final_results	[FINAL PAGE/LINES: Cohorts]	Primary dataset descriptors are cited; the present study performs no new annotation and reports only source-available annotation details.
C17	Describe annotation of the testing dataset	not_applicable	[FINAL PAGE/LINES: External Cohort]	No new test annotation is performed; processed BraTS-Africa reference labels are used under source terms.
C18	Report inter- and intrarater variability or mitigation	no	[FINAL PAGE/LINES: Limitations]	Rater-level repeat annotations are unavailable; this is retained as a limitation and no variability estimate is invented.
C19	Describe assignment to training, tuning, and testing partitions	yes_pending_final_page_lines	[FINAL PAGE/LINES: Study Design and External Evaluation]	splits/q1q2_v2/split_metadata.json; splits/q1q2_v2/external_test.csv.
C20	State the level at which partitions are disjoint	yes_pending_final_page_lines	[FINAL PAGE/LINES: Study Design and Leakage Controls]	Patient-level disjointness, cross-edition identity audit, and external overlap audit are explicit.
C21	Explain the intended testing sample size	yes_pending_final_page_lines	[FINAL PAGE/LINES: Statistical Analysis]	Fixed public external cohort with precision/power planning; reports/q1q2_v2/external_precision_analysis.json.
C22	Describe model inputs, outputs, layers, and modifications in reconstructable detail	yes_pending_final_page_lines	[FINAL PAGE/LINES: Models; Supplementary Table S3]	configs/q1q2_v2/model_matrix.yaml and executable implementations.
C23	Report software libraries, frameworks, packages, and versions	yes_pending_final_page_lines	[FINAL PAGE/LINES: Reproducibility]	environment/q1q2_v2-environment.json; environment/q1q2_v2-requirements-lock.txt.
C24	Describe parameter initialization and starting weights	yes_pending_final_page_lines	[FINAL PAGE/LINES: Models and Training]	Run-resolved model initialization and pretrained-weight status are recorded by configuration and metadata.
C25	Describe training, augmentation, optimization, stopping, and hyperparameters	yes_pending_final_page_lines	[FINAL PAGE/LINES: Loss Selection and Optimization]	configs/q1q2_v2/training_protocol.yaml; best and terminal checkpoints are required.

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C26	Describe selection of the final model or models	yes_pending_final_page_lines	[FINAL PAGE/LINES: Loss Selection and External Evaluation]	Development-only loss/checkpoint rules and the no-external-selection guard are explicit.
C27	Describe ensembling	yes_pending_final_page_lines	[FINAL PAGE/LINES: External Evaluation]	Fold-seed checkpoint predictions use a prespecified strict-majority nested-region aggregation.
C28	Define performance metrics and their task relevance	yes_pending_final_page_lines	[FINAL PAGE/LINES: Central Evaluation]	configs/q1q2_v2/evaluation.yaml; regional, boundary, surface, and lesion endpoints.
C29	Describe statistical uncertainty and comparison methods	yes_pending_final_page_lines	[FINAL PAGE/LINES: Statistical Analysis]	configs/q1q2_v2/statistical_analysis_plan.yaml; paired and hierarchical uncertainty with multiplicity control.
C30	Report robustness or sensitivity analyses	partial_pending_final_results	[FINAL PAGE/LINES: Results and Supplement]	Evaluation thresholds, connectivity, HD95 handling, budget, loss interaction, and external subgroup sensitivities are prespecified.
C31	Describe and validate explainability methods, if used	not_applicable	[FINAL PAGE/LINES: Methods or checklist note]	No saliency or post hoc explainability method is used or claimed.
C32	Report evaluation on internal data	yes_pending_final_page_lines	[FINAL PAGE/LINES: Development Results]	Patient-level cross-validation is reported; the previously opened legacy internal subset is not reused for v2 inference.
C33	Report testing on external data	partial_pending_final_results	[FINAL PAGE/LINES: External Evaluation and Results]	Gate H is pending; configs/q1q2_v2/gate_h_external.yaml prohibits tuning and requires one audited session.
C34	Provide clinical-trial registration when applicable	not_applicable	[FINAL PAGE/LINES: Study Design]	Retrospective public-data methodology study; no prospective clinical trial is conducted.
C35	Report included and excluded patient counts	partial_pending_final_results	[FINAL PAGE/LINES: Cohort flow and Results]	Counts are claim-bound and the final flow diagram is generated from manifests.
C36	Report demographic and clinical characteristics of each dataset and partition	partial_pending_final_results	[FINAL PAGE/LINES: Cohort table and external subgroups]	All source-available characteristics and explicit missingness are reported; absent demographics are not imputed.

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C37	Report model performance and statistical uncertainty on all relevant datasets	partial_pending_final_results	[FINAL PAGE/LINES: Results and Supplement]	Final Gate J values are pending; every frozen model and failed outcome must remain visible.
C38	Report diagnostic-performance estimates and precision for classification tasks	not_applicable	[FINAL PAGE/LINES: checklist note]	The task is semantic/lesion segmentation, not patient-level diagnostic classification; segmentation uncertainty is addressed under C29 and C37.
C39	Provide failure analysis and examples of incorrect results	partial_pending_final_results	[FINAL PAGE/LINES: Lesion and Qualitative Results]	Prespecified post-evaluation selection rules include hard, false-positive, boundary, and disagreement cases.
C40	Discuss study limitations	yes_pending_final_page_lines	[FINAL PAGE/LINES: Limitations]	Public retrospective data, annotation uncertainty, one external cohort, system differences, MPS scope, and reproduction boundaries are explicit.
C41	Discuss implications for practice and the intended clinical role	yes_pending_final_page_lines	[FINAL PAGE/LINES: Discussion and Conclusion]	The manuscript states that clinical utility and universal transportability are not established.
C42	Provide the full protocol or additional technical details	yes_pending_final_page_lines	[FINAL PAGE/LINES: Reproducibility and Supplement]	Versioned configs, supplement, code, and gate documentation provide technical detail.
C43	State availability of software, model, and data	partial_pending_final_results	[FINAL PAGE/LINES: Data and Code Availability]	Code is versioned; raw data remain under source terms; final release/checkpoint availability awaits Gate K and author confirmation.
C44	State funding, support, and funder role	no	[FINAL PAGE/LINES: Funding]	submission/ AUTHOR_CONFIRMATION_REQUIRED.md; no funding statement is inferred from repository artifacts.

Finalization rule

The PDF may be produced only after the result-bearing manuscript is rendered, the final document is paginated, every Yes/Partial row has a real page/line reference, unresolved result tokens are absent, and the authors confirm all declaration-dependent entries.